IP Addressing & Subnetting

1.List the IP address classes (A, B, C, D, E) and give one real-world example of a device or service that might use each class. Write your answers in a table format.

ANS:

IP Address Classes (A, B, C, D, E)

Class	IP Address Range	Use	Real-Life Example
A	1 – 126	Used for very big networks.	Large companies.
B	128 – 191	Used for medium networks.	Colleges and universities.
C	192 – 223	Used for small networks.	Home Wi-Fi router.
D	224 – 239	Used to send data to many devices at once.	Live video streaming.
E	240 – 255	Used for testing and research.	Network experiments.

2.Given the IP address 192.168.10.45 with subnet mask 255.255.255.0, calculate the network address, broadcast address, and the range of valid host addresses.

Hint: Show your working steps for each calculation.

ANS:

IP Address Calculation

Given:

- IP Address: 192.168.10.45
- Subnet Mask: 255.255.255.0

Step 1: Network Address

Change the last number to 0.

Answer: 192.168.10.0

Step 2: Broadcast Address

Change the last number to 255.

Answer: 192.168.10.255

Step 3: Valid Host Range

- First IP: 192.168.10.1
- Last IP: 192.168.10.254

Final Answer

Details	**Answer**
Network Address	192.168.10.0
Broadcast Address	192.168.10.255
First Host	192.168.10.1
Last Host	192.168.10.254
Total Devices	254

3.Create a subnetting plan for a small food delivery startup like Zomato that needs 4 separate subnets: one for Delivery Partners, one for Restaurants, one for Customers, and one for Admins. Assign a unique subnet (with network address and subnet mask) for each group using the 10.0.0.0/24 network.

ANS:

Subnetting Plan for Food Delivery Startup

We have the network 10.0.0.0/24 and need 4 separate subnets.

We divide it into 4 equal parts.

No.	**Group**	**Network Address**	**Subnet Mask**
1	Delivery Partners	10.0.0.0	255.255.255.192 (/26)
2	Restaurants	10.0.0.64	255.255.255.192 (/26)
3	Customers	10.0.0.128	255.255.255.192 (/26)
4	Admins	10.0.0.192	255.255.255.192 (/26)

Explanation (Easy Words)

1. Delivery Partners – Used by delivery boys.

2. Restaurants – Used by restaurant staff.
3. Customers – Used by people ordering food.
4. Admins – Used by company staff.

4.You are given the following scenario: A college WiFi network uses the IP range 172.16.0.0/16. The admin wants to create at least 20 subnets for different buildings. Calculate the new subnet mask and the number of hosts available per subnet.

Constraint: Show your calculation steps and final answers.

ANS:

Subnetting Calculation

Given:

- IP Address: 172.16.0.0/16
- Required subnets: 20

Step 1: Calculate Subnet Bits

We need at least 20 subnets.

- \(2^4 = 16\) (Not enough)
- \(2^5 = 32\) (Enough)

So, we borrow 5 bits.

Step 2: Find New Subnet Mask

- Old subnet mask: 255.255.0.0
- New subnet mask: 255.255.248.0
- New CIDR: /21

Step 3: Calculate Hosts

- Remaining host bits = 11
- \(2^{11} - 2 = 2046\)

So, each subnet can have 2046 devices.

Final Answer

Details	Answer
Original Network	172.16.0.0/16

Required Subnets	20
New Subnet Mask	255.255.248.0
New CIDR	/21
Total Subnets	32
Hosts per Subnet	2046

5.A friend sends you a message asking why their device with IP 192.168.1.300 is not connecting to the network. Write a short explanation identifying the error and what a valid IP address in that subnet should look like.

ANS:

P Address Error

The IP address 192.168.1.300 is wrong because 300 is too large.

Each IP address number must be between 0 and 255.

Correct IP examples:

- 192.168.1.1
- 192.168.1.10
- 192.168.1.100